



Operational Amplifier GAIN AND DC OFFSET PROGRAM

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Presented here is a program to find the gain and DC offset values of an operational amplifier (or op-amp). While doing practical electronics experiments, especially in analogue electronics, we come across a very versatile component, that is, the op-amp. It is a low-cost single-chip solution to clumsy and difficult-to-troubleshoot multiple-transistors based designs.

A single op-amp can be interfaced to any sensor or transducer, and with proper biasing and power supply it may produce desired output signal levels. But op-amps, too, need external biasing with the use of minimal resistances and reference voltages.

Basically, a designer faces the challenges to set the gain and DC offset of the op-amp for producing the required output. There is a set of relationships that can be incorporated to find those required biasing components.

This software program does the work for you automatically by taking the required parameters from you, such as reference voltage, required output and input voltage swings. It then outputs a

recommended design. You need to choose at least one component value by referring the datasheet of the op-amp to calculate the rest automatically. Then, you need to find the nearest component values to incorporate in the design.

To design such a circuit, there should be some known values that you need to feed as inputs in the program. For example, parameters like voltage level of a stable reference (V_{ref}), full-scale output voltage (V_{outfs}), zero-scale output voltage (V_{outzs}), full-scale input voltage (V_{infs}) and zero-scale input voltage (V_{inzs}) are required.

Software

The software program is written using HTML, PHP and JavaScript, and can be incorporated in a system having PHP installed. It runs under PHP development environment. HTML is used as front-end software, whereas PHP is used as back-end software. JavaScript file checks for blank fields in HTML form.

PHP is incorporated in a Web server. So multiple user clients within the network can use this software simultaneously, like in a lab or college. The IDE used to develop this program is NetBeans, having PHP environment.

Three programs used in this project are opampRequirementForm.html, opampCircuit.php and opampDesign.js.

When you run opampRequirementForm.html, you get a window as shown in Fig. 1. This is the input form, where you need to enter all input parameters.

Fig. 1: OpampRequirementForm.html program output window

Fig. 2: Positive gain and positive offset